EE 474 Introduction to Control Systems Homework 7

Due Tuesday October 23, 2023

Reading:

- Franklin *et al.*: Chapter 4, section 4.2 to end.

Homework :

1 Review questions 4.1 and 4.2

4.1 Give three advantages of feedback in control.
4.2 Give two disadvantages of feedback in control.

2 Text problem 4.6

Text problem 4.6 : Consider the DC motor control system with rate (tachometer) feedback shown in Fig. 4.26(a).

(a) Find values for K' and $k_t^{'}$ so that the system of Fig. 4.26 (b) has the same transfer function as the system of Fig. 4.26(a).

(b) Determine the system type with respect to tracking θ_r and compute the system K_v in terms of parameters K' and $k_t^{'}$.

(c) Does the addition of tachometer feedback with positive k_t, increase or decrease K_v ?

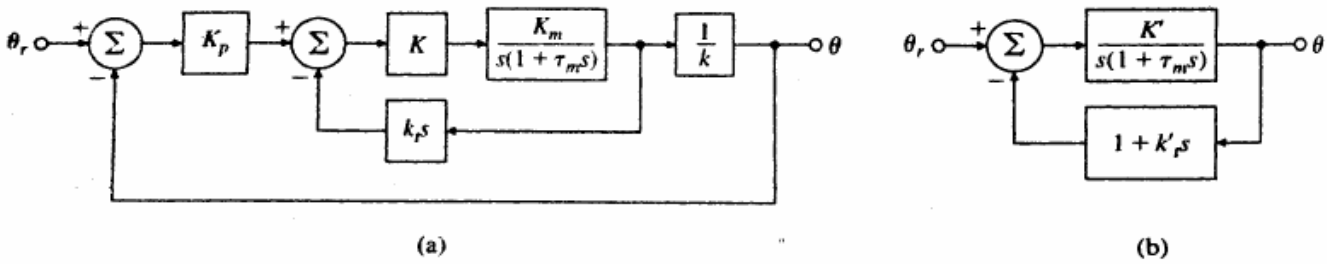


Figure 4.26
Control system for Problem 4.6

3 Working with a control loop and poles and zeros

For a system described by the block diagram of figure 1.

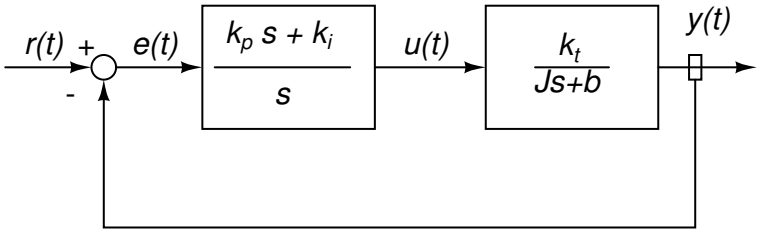


Figure 1: Block diagram of motor with control.

A) Form the transfer function $T_{ry}(s)$.

B) With parameters $k_p = 1.5$, $k_i = 0.4$, $k_t = 1.8$, $J = 15$, $b = 2.0$, determine the numerical values for the numerator and denominator of $T_{ry}(s)$.

C) What are the poles and zeros of $T_{ry}(s)$?

D) Based on the closed-loop poles, determine these four quantities. For each, write the expression that is used to calculate an approximate value from the pole locations.

D.1) T_r, the rise time

D.2) ρ, the damping factor

D.3) P.O., the percent overshoot

D.4) T_s, the settling time

E) In Matlab, setup the system object (example code below) Plot the step response, and directly measure the four quantities above. (Note, in Matlab you can left-click on the step-response curve to draw a dialog box with data at the selected point, give it a try).

```
%%  Example Matlab code (modify as desired)
kt = 1.8, J = 15, b = 2
kp = 1.5, ki = .4
Gc = tf( [kp, ki], [1 0]);                %% Setup TF objects
Gp = tf(kt, [J b]);
Try = feedback(Gc*Gp, 1)                  %% Make closed-loop TF
CLPoles =  pole(Try)
alpha  = -real(CLPoles(1))
omega  = imag(CLPoles(1))
omegan = norm(CLPoles(1))
rho = alpha/omegan                        %% damping factor
PO = 100* exp( -(pi*rho)/sqrt(1-rho^2))   %% percent overshoot

stepinfo(Try)       %% Try this also, stepinfo
                    %% processes information from the step response
```

F) In parts D and E you computed performance characteristics in two ways,

a) by location of the dominant poles, and

b) by computing and measuring the step response

These two approaches are characterized by

i) One approach is approximate, but provides equations that can be used for design

ii) One approach is exact, buy applies only to that specific system, and generally does not provide design insight.

Which method has which characteristic ? Match each of methods *a* and *b*, to characterizations *i* and *ii*

a _____, b _____

G) Now change parameters to $k_p = 5.0$ and $k_i = 1.2$.

What are the new closed-loop pole locations ?

Make a plot of the S-plane. Plot the original closed-loop pole locations (from part C), and the new closed-loop pole locations.

H) Without calculating any quantities, but looking at how the poles moved, and considering the analysis equations (in part D),

H.1) Do you expect the rise time to be faster or slower ?

H.2) Do you expect the damping factor to be greater or less ?

H.3) Do you expect the percentage overshoot to be greater or less ?

H.4) Do you expect the settling time to be greater or less ?

I) In Matlab, setup the new $G_c(s)$ and $T_{ry}(s)$ as in part E, and run

```
>> stepinfo(Try)
```

How did the values of T_r, ρ, PO, T_s change relative to what you got in part E ?

Did the changes agree with your expectations ?

4 Text problem 4.7

Text problem 4.7 : A block diagram of a control system is shown in Fig. 4.27.

(a) If r is a step function and the system is closed-loop stable, what is the steady-state tracking error?

(b) What is the system type?

(c) What is the steady-state error to a ramp velocity 5.0 if $K_2 = 2$ and K_1 is adjusted to give a system step overshoot of 17% ?

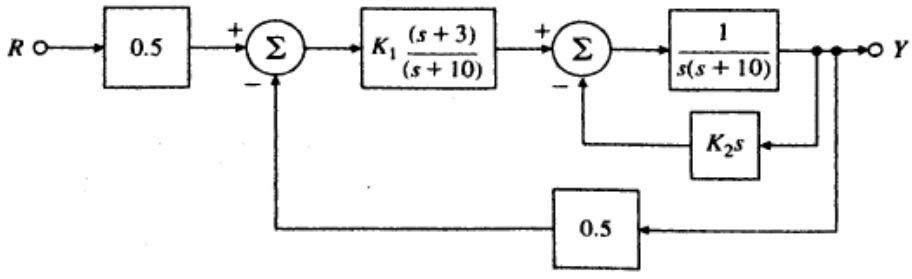


Figure 4.27
Closed-loop system for Problem 4.7